

Numerical Analysis PhD Exam, January 2009

Answer any 8 questions.

1. Let $M = \begin{bmatrix} A & B^t \\ B & C \end{bmatrix}$ be real symmetric and positive definite matrix. Let $S = C - BA^{-1}B^t$ be the Schur complement. Prove that S is symmetric and positive definite.
2. Let $\|\cdot\|_p$ for $1 \leq p \leq \infty$ denote the norm on $m \times n$ matrices induced by the ℓ^p -norm on vectors in $\mathbb{C}^m$ and $\mathbb{C}^n$. Prove that

$$\|A\|_2 \leq \sqrt{\|A\|_1 \|A\|_\infty}, \quad \text{for all } A \in \mathbb{C}^{m \times n}.$$

3. Show that if P and Q be Hermitian positive definite matrices satisfying $x^*Px \leq x^*Qx$, for all $x \in \mathbb{C}^n$, then $\|P\|_F \leq \|Q\|_F$.
4. Let T be any square matrix and let $\|\cdot\|$ denote any induced norm. Prove that $\lim_{n \rightarrow \infty} \|T^n\|^{1/n}$ exists and equals $\inf_{n=1,2,\dots} \|T^n\|^{1/n}$.
5. Let P and Q be two $m \times m$ orthogonal projectors. Prove that $\|P - Q\|_2 \leq 1$.
6. Let $x_0, x_1, \dots, x_n$ be distinct points in a finite interval $[a, b]$ and $f \in C^1[a, b]$. Show that for any given $\varepsilon > 0$ there exists a polynomial p such that $\|f - p\|_\infty < \varepsilon$ and $p(x_i) = f(x_i)$ for all $i = 0, 1, \dots, n$ (where $\|\cdot\|_\infty$ denotes the $L^\infty(a, b)$ -norm).
7. Let $\hat{f}(s)$ be the continuous Fourier transform of $f(t) \in L^2[-\infty, \infty]$. Suppose further that $\hat{f}(s) = 0$ for $|s| > \pi$. Derive the interpolation formula

$$f(t) = \sum_{k=-\infty}^{\infty} f(k) \frac{\sin(\pi(t-k))}{\pi(t-k)}.$$

8. Let $L_n(f)$ be the Lagrange polynomial interpolating a function f at nodes $a = x_0 < x_1 < \dots < x_n = b$.
 - (a) Give the formula for $L_n(f)$.
 - (b) Prove that there is a unique polynomial of degree at most n interpolating f at the nodes.
 - (c) State and prove the formula for the error $f(x) - L_n(f)(x)$ when $f(x) \in C^{n+1}[a, b]$.
9. Assume that g is continuously differentiable real-valued function and $a \leq g(x) \leq b$ on $[a, b]$. Show the following:
 - (a) There is an $\alpha \in [a, b]$ such that $g(\alpha) = \alpha$.
 - (b) If $|g'(x)| < 1$ on $[a, b]$, then there is only one fixed point on the interval $[a, b]$.
 - (c) If we generate iterates using the recurrence $x_{n+1} = g(x_n)$ starting from some $x_0 \in [a, b]$, then we have

$$|\alpha - x_n| \leq \frac{\lambda^n}{1 - \lambda} |x_1 - x_0| \quad \text{where} \quad \lambda = \max_{x \in [a, b]} |g'(x)|.$$

10. Let $\{\phi_n(x) : n \geq 0\}$ be an orthogonal family of polynomials on the interval (a, b) with weight function $w(x) \geq 0$ for all $x \in [a, b]$. Show that the polynomial $\phi_n(x)$ has exactly n distinct real roots in the open interval (a, b) .